Creating a Prototype Visualization Tool for Analyzing Pre-Pubescent Lupus Data

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Abstract

Systemic Lupus Erythematosus (SLE), commonly known as Lupus, is a chronic autoimmune disease, that can target any organ in the body. The disease has been reported to be most dangerous for adolescents in the first year after diagnosis. The aim of this research is to develop a prototype visualization tool for analyzing pre-pubescent Lupus data. This research utilizes data visualization techniques and visual analytics to examine a variety of variables that track Lupus symptoms and puberty progression. This will enable incremental refinement of the interface and assess functionality of the tool. The current results show the earlier an adolescent is in puberty, the worse the Lupus symptom flares are. Theoretically, this could be due to the intense surge of hormones into the body, primarily leptin. The results suggest if the correct hormone can be identified as the catalyst for the flares, the hormone could be suppressed to slow down the progression of puberty to provide improved quality of life for SLE adolescents. Continued work will include adding statistical elements to the backend to provide summary information about the data to help users better understand the data. To enable this functionality, the following features will be integrated into the tool: creating a filter for variables, a breakdown menu for each individual patient, and a cross-variable analysis.

What is Lupus?

Figure 1 SLE Butterfly Rash [6]

SLE is an autoimmune disease generally diagnosed between the ages of 15 and 45. About 5,000,000 cases of the disease have been diagnosed and nearly 90% of them are Women [5]. There is currently no cure for the disease, just treatments. Some symptoms include intense skin rashes, often on the face in the shape of a butterfly (as illustrated Figure 1), possible kidney failure, and pain, stiffness, and swelling of the joints. People with the condition are also at risk for diseases such as Osteoporosis and Fibromyalgia. In addition to having a negative impact on a person's life, it can also prove deadly. There are large amounts of misdiagnosis and little is known about what causes the disease. [3]

Methodology

- Determined what variables took priority for analysis based on Dr. Kathleen O'Neil's request
- Create multiple prototype visualizations to determine what was possible for tool creation and what should be used.
- After using software such as Cytoscape and Gapminder, it was decided to code a tool from scratch using Python as it would allow the data to stay private
- Decided upon the coding library Bokeh, as it is a good visualization software that works similarly to web-based visualization tools
- The SLE data was to fit in a pre-existing visualization
- Began using the pre-existing visualization as a basis for the code for the prototype

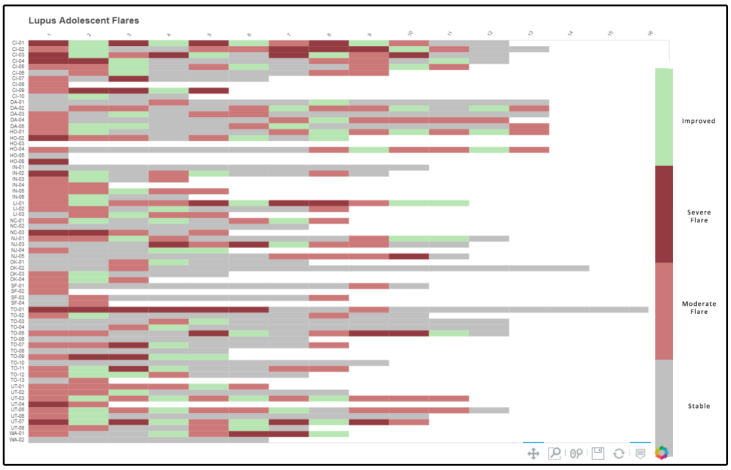

Figure 2 Visual representation of Lupus Adolescent Flares

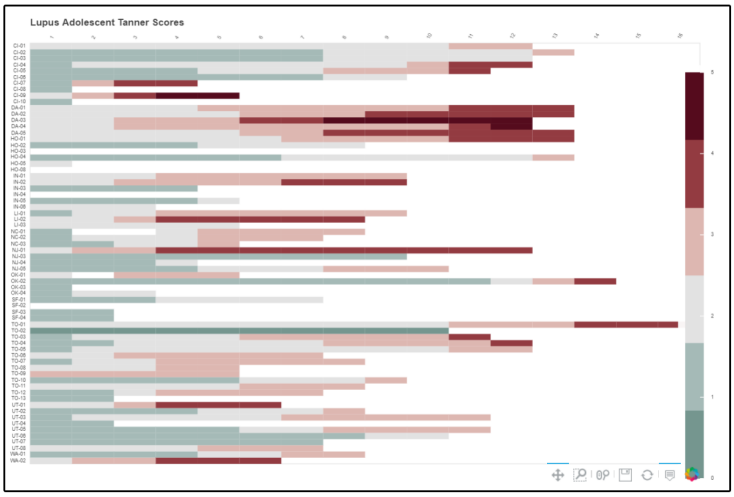

Figure 3 Visual representation of Lupus Adolescent Tanner Scores

Tested Variables

Tanner Score: A measurement scale from 1 to 5 that is meant to determine how far along someone is in the puberty process. 1 means puberty has yet to begin, and 5 means puberty is complete [4]

Leptin: Leptin is a fat hormone that is crucial to starting puberty. When puberty begins it surges, but once that process is complete, girl's leptin count stays high while a boys drops, but is still above where they started. [4]

Lupus Flares: A lupus flare is a period in which someone with SLE has an increased number of symptoms relative to the Systematic Lupus Erythematosus Disease Activity Index (SLEDAI). This is more commonly referred to as the SLEDAI Score. [2] [4]

Moderate Flare: A rise in SLEDAI score by more than 3 if the total SLEDAI score is less than 12. [1]

Severe Flare: A SLEDAI score that is 12 or greater. [1]

Results

Figures two and three are the results of the python visualization where figure two represents the SLE flares in the patients and figure 3 represents the patient's progression through puberty via tanner stage. Figure 2 can only have four different values: gray means the patient is stable, light red means there is a moderate flare, dark red is a severe flare, and green represents improvement from a flare in the previous visit. Comparing the two figures, it appears as though a patient has flares when they just begin puberty as when the flares occur, patients are in stages one or two of puberty. This would mean that as the fat hormone leptin surges at the start of puberty, the influx may cause the body to overreact and increase symptoms. While there could be a variety of hormones responsible for the increase of symptoms, it is still likely that the hormones are to blame. If this is the case, then hormone suppressors may assist in alleviating the severity of the adolescent's symptoms by slowing down the rate in which they are introduced to the body.

Future Work

After having made an initial prototype with multiple views of different visualizations, the next step is to create a filter that connects all these different views within the heatmap. Furthermore, statistical analysis will be added to the backend to provide summary information about the data. After this, a breakdown view will be created to look at different variables for each of the individual patient's week, by week. There are also plans in place to bring on another student to assist in creating a cross analysis feature to give the user of the tool more freedom when it comes to changing the different variables within the heatmap and other planned visualizations. The end goal is to create a tool that assists doctors in analyzing various autoimmune diseases, based on the framework created with our SLE visualization tool.

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